

normxcorr2 Function Test Results

Test Case 1: Basic Matching Template

Template:

1. 2.

3. 4.

Search Image:

0. 0. 0.

1. 2. 0.

3. 4. 0.

Output:

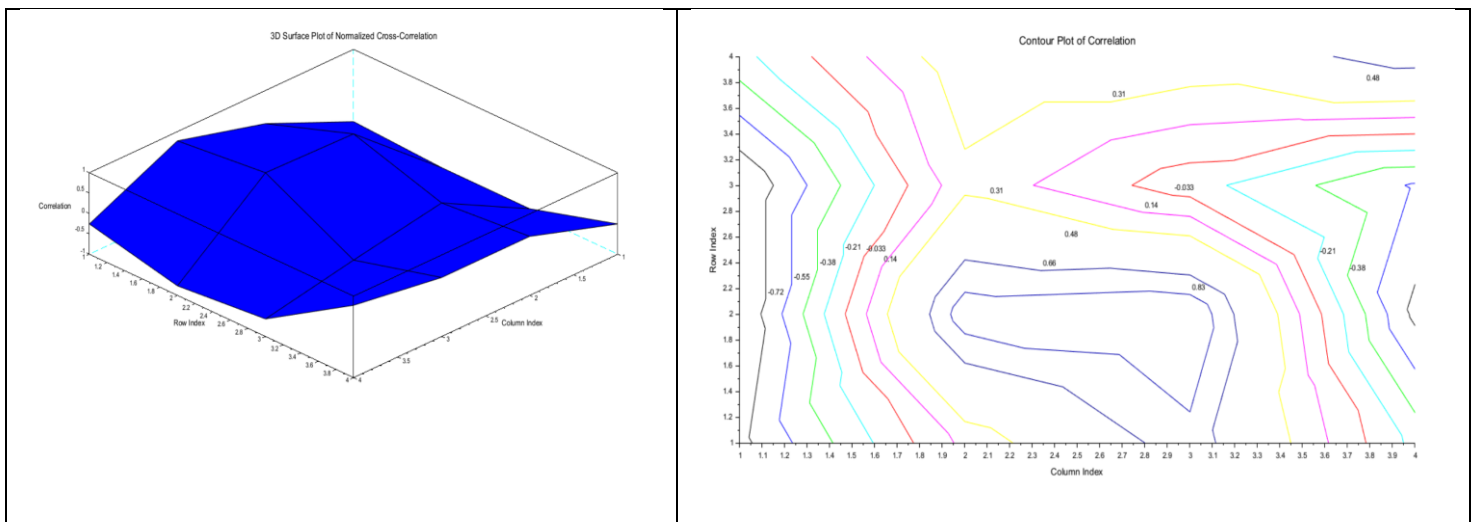
-0.7745967 -0.8944272 -0.8944272 -0.2581989

0.1860886 0.9438798 0.2581989 0.4472136

0.7734534 1. -0.13484 0.4472136

-0.2581989 -0.7678912 -0.5698413 0.7745967

Visualization Plots



Test Case 2: Rectangular Template Matching

Template:

2. 4. 1.

7. 3. 5.

Search Image:

1. 0. 2. 1. 3.

4. 2. 4. 1. 0.

7. 3. 5. 2. 1.

2. 6. 1. 4. 5.

3. 1. 2. 7. 6.

Output:

-0.3023716 -0.1876913 -0.4285229 -0.8811878 -0.1923418 -0.7553101 0.7559289

0.6240886 0.3310325 0.5971484 0.225877 0.1048285 -0.0872638 -0.7618367

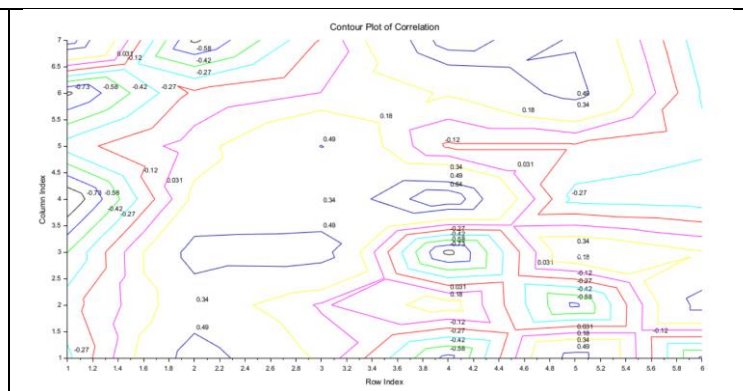
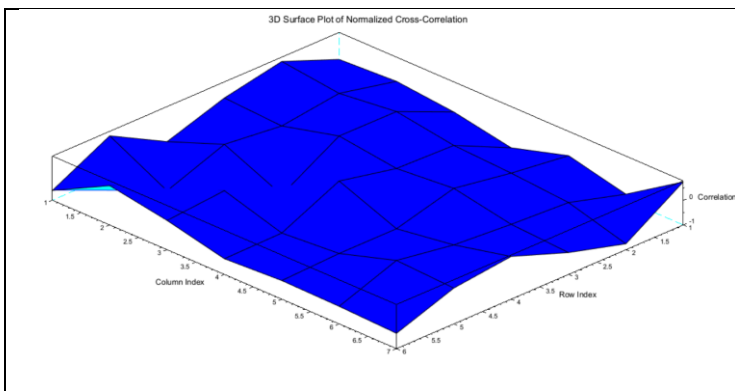
0.144911 0.0104979 0.609186 0.2935198 0.4929336 0.0346387 -0.110125

-0.6299642 0.2693705 -0.7824608 0.7917947 -0.146385 0.2032924 0.7412888

0.6180859 -0.6158011 0.3958973 -0.321908 -0.1465365 0.5802197 0.459759

-0.6047432 0.6023754 0.1923418 -0.3614647 -0.2433081 -0.2849859 -0.3779645

Visualization Plots



Test Case 3: Floating Point Inputs

Template:

0.5 1.5

2.5 3.5

Search Image:

1.1 2.2 3.3

4.4 5.5 6.6

7.7 8.8 9.9

Output:

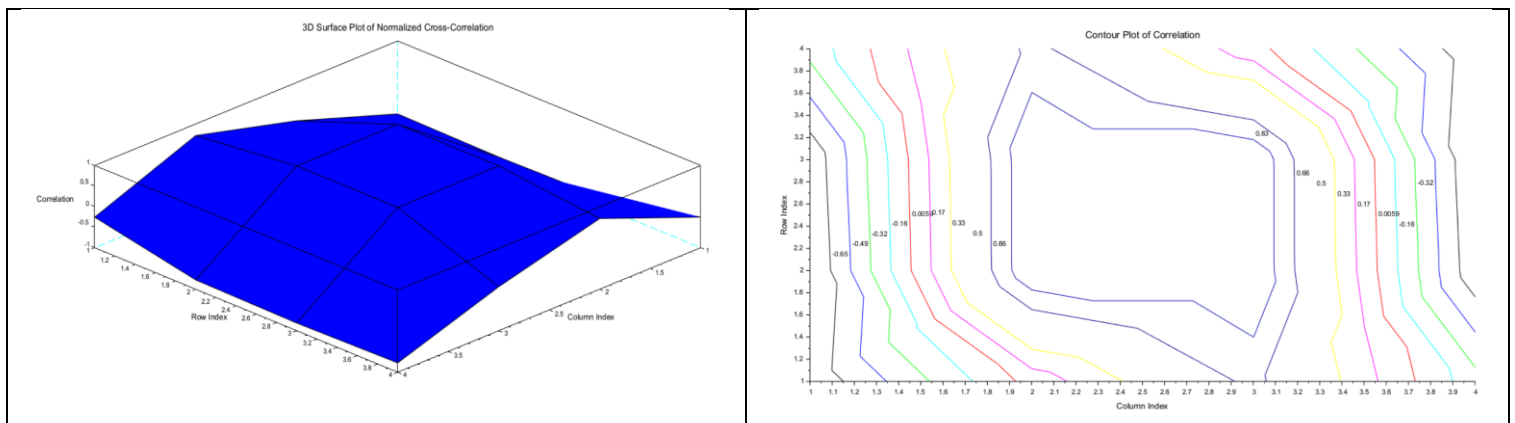
-0.7745967 -0.8140916 -0.7745967 -0.2581989

0.0681994 0.9899495 0.9899495 0.7181848

0.7181848 0.9899495 0.9899495 0.0681994

-0.2581989 -0.7745967 -0.8140916 -0.7745967

Visualization Plots



Test Case 4: Negative Values

Template:

-1. -2.

3. 4.

Search Image:

-4. -3. -2.

-1. 0. 1.

2. 3. 4.

Output:

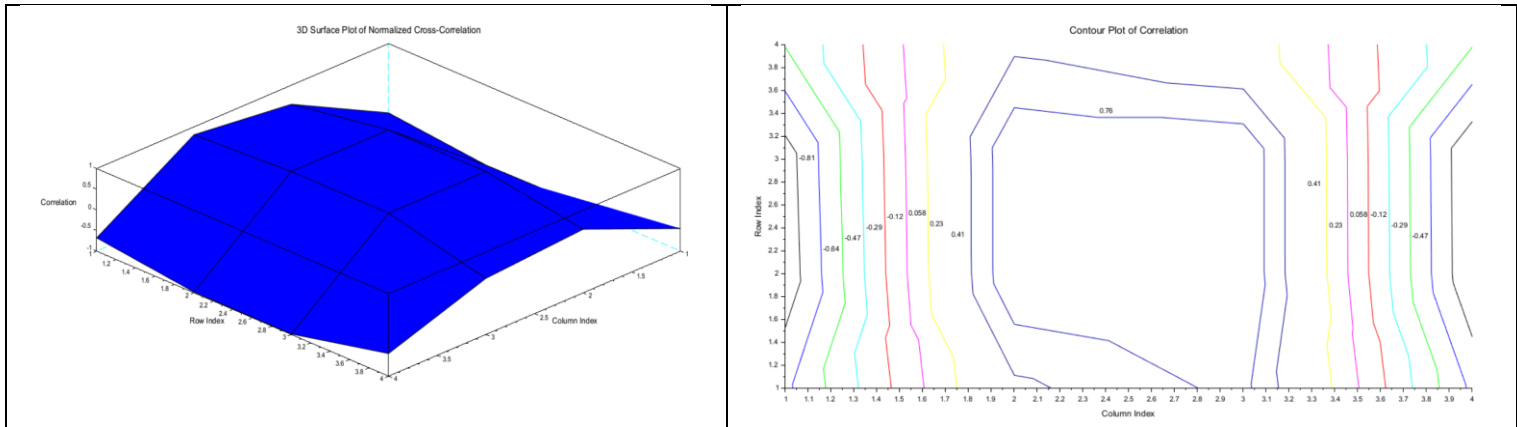
-0.6793662 -0.9336996 -0.9058216 -0.4529108

0.5383337 0.9302605 0.9302605 0.5399055

0.8098583 0.9302605 0.9302605 0.3588892

-0.6793662 -0.9813068 -0.9886231 -0.4529108

Visualization Plots



Test Case 5: Single Pixel Template

Template:

5.

Search Image:

1. 2. 3.

4. 5. 6.

7. 8. 9.

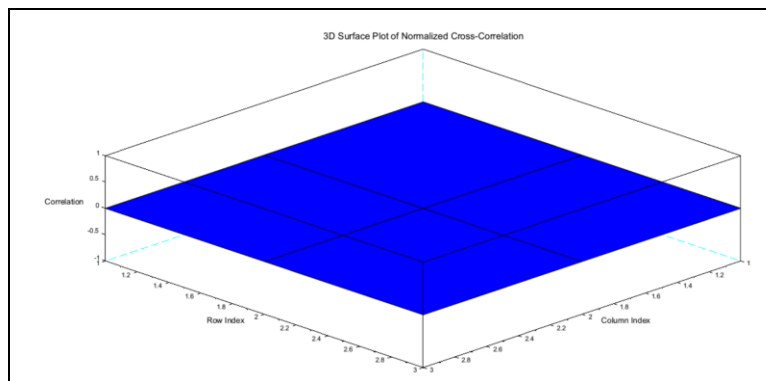
Output:

0. 0. 0.

0. 0. 0.

0. 0. 0.

Visualization Plot



Test Case 6: Template Equals Image

Template:

2. 1.
4. 3.

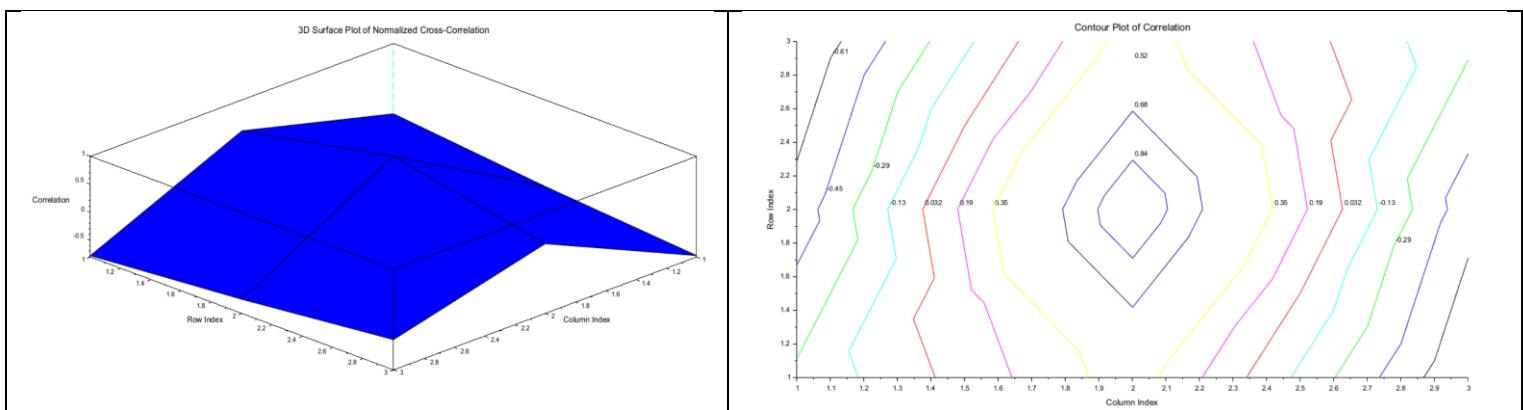
Search Image:

2. 1.
4. 3.

Output:

```
-0.2581989 -0.5477226 -0.7745967
 0.4472136 1.          0.4472136
-0.7745967 -0.5477226 -0.2581989
```

Visualization Plots



Test Case 7: Template Larger Than Image

Template:

1. 2. 3.
4. 5. 6.
7. 8. 9.

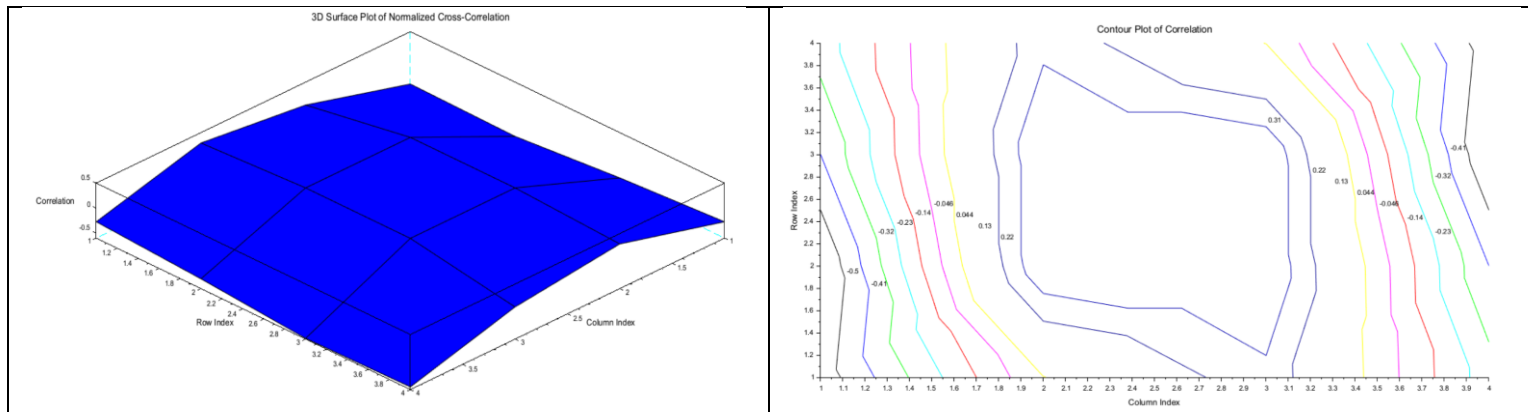
Search Image:

1. 2.
3. 4.

Output:

-0.5477226	-0.5852927	-0.4052027	-0.2738613
0.0417635	0.4041452	0.4041452	0.2923442
0.2923442	0.4041452	0.4041452	0.0417635
-0.2738613	-0.4052027	-0.5852927	-0.5477226

Visualization Plots



Test Case 8: Constant Template

Template:

1. 1.
1. 1.

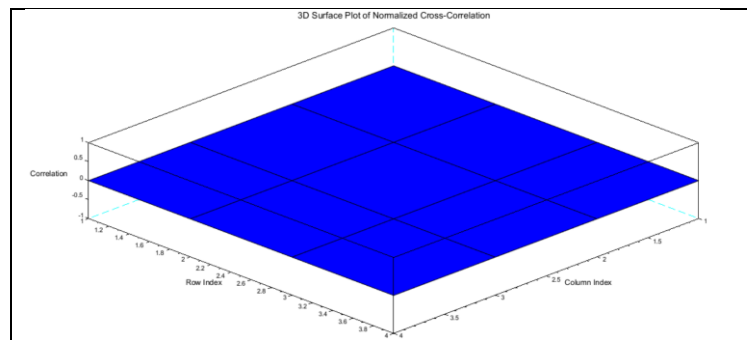
Search Image:

1. 2. 3.
4. 5. 6.
7. 8. 9.

Output:

0. 0. 0.
0. 0. 0.
0. 0. 0.

Visualization Plot



Test Case 9: Larger Image with Repeated Pattern

Template:

1. 2.
3. 4.

Search Image:

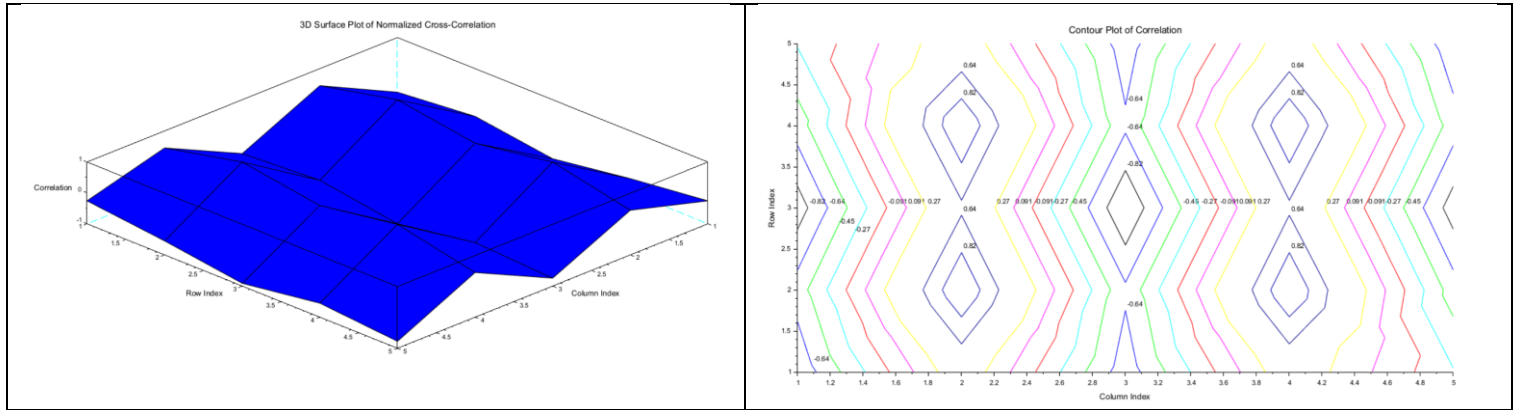
1. 2. 1. 2.
3. 4. 3. 4.
1. 2. 1. 2.
3. 4. 3. 4.

Output:

-0.7745967	-0.5477226	-0.9128709	-0.5477226	-0.2581989
0.4472136	1.	0.6	1.	0.4472136
-0.745356	-0.6	-1.	-0.6	-0.745356

0.4472136	1.	0.6	1.	0.4472136
-0.2581989	-0.5477226	-0.9128709	-0.5477226	-0.7745967

Visualization Plots



Test Case 10: Invalid Number of Arguments

Template:

1. 2.
3. 4.

Search Image: [not provided]

Output: "normxcorr2: Wrong number of input arguments."